

Context-Adaptive Empathy Scales for HRI: A Multi-Agent LLM Approach

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Team Information and Expected contributions

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1. Motivation

Empathy plays a foundational role in interactive AI systems, especially in human–robot interaction (HRI), where empathic behavior influences trust, cooperation, and long-term acceptance [3, 1]. Crucially, empathy in HRI is highly *contextual*: its expression—and users’ perception of it—varies widely across tasks, roles, and interaction settings. A collaborative robot in an assembly task may express empathy through adaptive task coordination (e.g., slowing down when a human hesitates), whereas a counseling-oriented conversational agent demonstrates empathy through reflective dialogue. A home service robot may instead convey empathy by anticipating user needs or adjusting behavior in response to frustration. Although all represent “empathy,” their forms differ sharply across platforms and use cases.

This variability makes measurement difficult. Existing instruments such as RoPE [4] and PETS [12] provide useful baselines but are either too specialized or too general to capture scenario-dependent differences. As a result, researchers lack tools that reflect how empathy *should* appear within a specific interaction context.

Developing scenario-specific empathy scales traditionally requires the full nine-step psychometric framework [2], including construct definition, item generation, expert review, pilot testing, and factor analysis. While rigorous, this process is slow, costly, and impractical to repeat for the diverse and rapidly expanding range of HRI scenarios.

Advances in large language models (LLMs) offer a potential path toward scalable solutions. LLMs can generate psychometric items [7, 8] and perform structured evaluation tasks such as UX analysis [6]. Multi-agent LLM systems such as ChatDev [11] and MetaGPT [5] further show that coordinated, role-specialized agents can emulate complex expert workflows. These developments motivate exploring whether similar architectures can support rapid creation of *scenario-sensitive* empathy scales for HRI.

This project aims to develop a lightweight, non-iterative multi-agent LLM prototype that generates scenario-specific empathy items and conducts preliminary evaluation using simulated “participants.” Comprehensive validation with human users is reserved for future work.

2. Background & Related Work

Empathy measurement in HRI. Prior studies highlight empathy’s importance for trust, acceptance, and interaction quality [13, 10]. RoPE [4] and PETS [12] represent important steps, but lack scenario sensitivity.

Scale development methodology. The widely adopted nine-step framework [2] requires expert review, item refinement, and multiple rounds of data collection; recent HCI work such as PETS[12] and PSC [9] follows this process. However, the required effort limits scalability.

LLMs for item generation. Prior work demonstrates that LLMs can produce coherent, high-quality assessment items and reasoning-based evaluations [7, 8, 6]. However, applications to HRI empathy remain unexplored.

Multi-agent LLM systems. Systems like ChatDev [11] and MetaGPT [5] show that role-specialized LLM agents can collaborate effectively, giving promise for automated scale development.

3. Problem Formulation & Expected Outcomes

Objective: Develop a multi-agent LLM system to generate scenario-sensitive empathy items and conduct initial LLM-based evaluation.

Research questions: (1) Can multi-agent LLMs generate coherent, scenario-specific empathy items? (2) Can LLM evaluation agents provide consistent preliminary judgments? (3) Do generated items appear more contextually appropriate than baseline scales such as PETS?

System components: **Construct Definition Agents** defines scenario-relevant empathy dimensions. **Item Generation Agents** produce diverse candidate items (6–12 per scenario). **Content Assessment Agents** refines wording and removes redundancy. **Evaluation Agents** simulate participant judgments of clarity, relevance, and scenario sensitivity.

The system will produce two to three scenario-specific empathy scales (e.g., robot teammate, home robot, counseling agent). No factor analysis or human user evaluation is included in this phase.

4. Planned Methods

The prototype will be implemented in Python using LLM APIs. Given a scenario description, the Construct Definition Agent extracts relevant constructs. Multiple Item Generation Agents independently generate candidate items to increase diversity. A Content Assessment Agent performs wording improvements and content consolidation. Evaluation Agents simulate human participants by providing Likert ratings and short rationales for each item. All evaluations are LLM-based and treated as feasibility signals rather than formal psychometric evidence.

No human data will be collected. Ethical considerations are limited, but any future real-user studies would require formal review.

5. Evaluation Plan

The system will be evaluated through LLM-based participant simulation. Evaluation agents will provide item-level clarity ratings, relevance ratings, scenario-sensitivity judgments, and concise justifications. Both SensEmoScale-generated items and baseline scales (PETS [12] and RoPE [4]) will be evaluated under identical procedures.

Metrics include mean clarity, contextual relevance, differences across scenarios, and stability across repeated simulation runs. Ablations will compare single-agent versus multi-agent generation, with versus without content refinement, and scenario-specific versus generic prompting. The evaluation is preliminary and does not constitute psychometric validation; full statistical analysis and user studies remain future work.

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